

UNIT-1

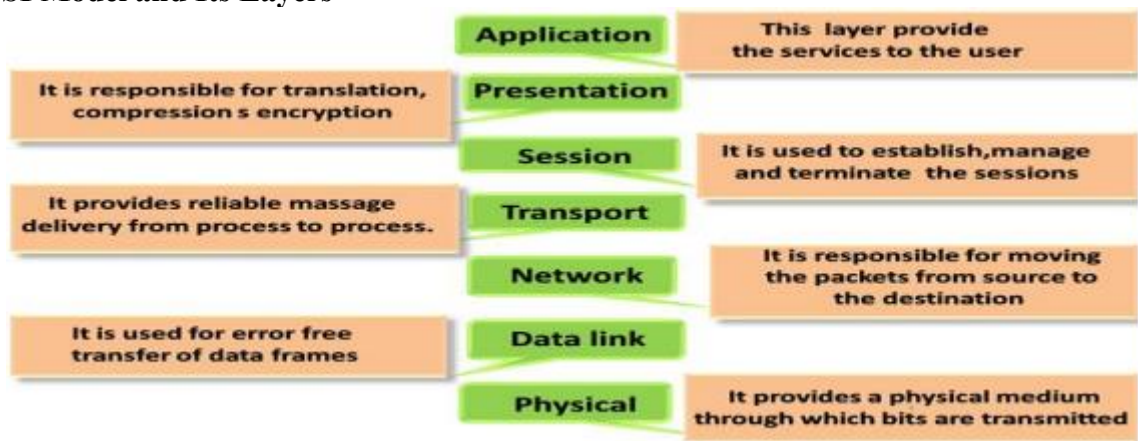
1. DEFINE COMPUTER NETWORKS AND EXPLAIN OSI LAYERS

Definition of Computer Networks

A **computer network** is a set of devices (called nodes) connected by communication links. A node can be a computer, printer, or any other device capable of sending or receiving data. Two computers are said to be interconnected if they can exchange information, whether through copper wire, fiber optics, microwaves, infrared, or satellites.

A computer network is a group of interconnected computers that share resources and communicate using a common technology.

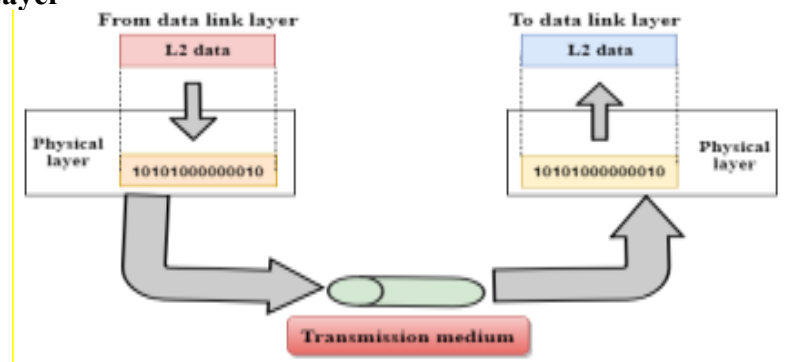
OSI Model and Its Layers



The **OSI (Open Systems Interconnection)** model is a reference model developed by ISO in 1984. It describes how information from a software application in one computer moves through a network to reach another computer.

It divides the networking process into **seven layers**, each with specific functions:

1. Physical Layer

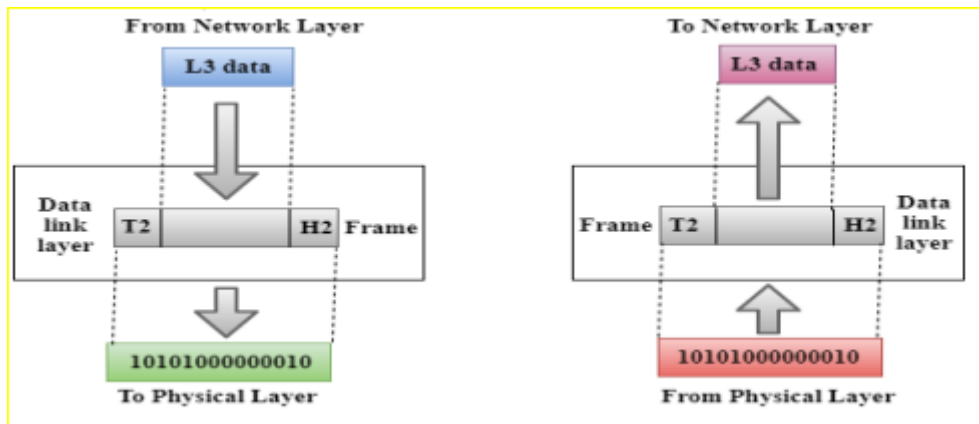


- Transmits individual bits from one node to another.
- Lowest layer of the OSI model.
- Establishes, maintains, and deactivates physical connections.
- Specifies **mechanical, electrical, and procedural** network interface specifications.

Functions of the Physical Layer:

- **Line Configuration:** Defines how devices connect physically.
- **Data Transmission:** Defines transmission mode (simplex, half-duplex, full-duplex).
- **Topology:** Defines how network devices are arranged.
- **Signals:** Determines the type of signal used for transmission.

2. Data-Link Layer



- Ensures **error-free transfer** of data frames.
- Defines the format of data on the network.
- Provides reliable and efficient communication between devices.
- Responsible for unique identification of devices in a local network.

Sub-layers:

1. **Logical Link Control (LLC):** Transfers packets to the Network layer, identifies protocol addresses, provides flow control.
2. **Media Access Control (MAC):** Controls how packets are placed on the physical medium.

Functions of Data-Link Layer:

- **Framing:** Converts bit streams into frames (adds headers/trailers).
- **Physical Addressing:** Adds destination address in the frame header.
- **Flow Control:** Maintains a constant data rate to avoid corruption.
- **Error Control:** Uses **CRC** (Cyclic Redundancy Check); retransmits if errors occur.
- **Access Control:** Decides which device controls the shared channel at a time.

3. Network Layer

- Manages device addressing and tracks device locations.
- Determines the **best path** for data transfer.
- Responsible for **routing and forwarding** packets.
- **Routers** operate at this layer.
- Protocols: **IP, IPv6**.

Functions of Network Layer:

- **Internetworking:** Provides logical connection between devices.
- **Addressing:** Adds source and destination IP addresses.
- **Routing:** Selects the optimal path to destination.
- **Packetizing:** Converts messages into packets (via IP).

4. Transport Layer

- Ensures **complete and reliable data transfer**.
- Transmits messages in correct order with no duplication.
- Splits data into **segments** at the sender and reassembles at the receiver.
- Provides **end-to-end communication** between source and destination.

Protocols:

- **TCP:** Reliable, connection-oriented, ensures ordering and error-checking.
- **UDP:** Unreliable, connectionless, faster, no acknowledgments.

Functions of Transport Layer:

1. **Service-point addressing:** Uses **port numbers** to deliver data to correct process.
2. **Segmentation & Reassembly:** Splits and reassembles messages with sequence numbers.
3. **Connection Control:** Provides connection-oriented and connectionless services.

4. **Flow Control:** End-to-end control to avoid congestion.
5. **Error Control:** Ensures error-free delivery through retransmission.

5. Session Layer

- Establishes, maintains, and synchronizes interaction between devices.

Functions of Session Layer:

1. **Dialog Control:** Manages sessions (half-duplex or full-duplex communication).
2. **Synchronization:** Adds checkpoints so that transmission can resume from last checkpoint if errors occur.

6. Presentation Layer

- Concerned with **syntax and semantics** of exchanged information.
- Acts as a **translator** between application and network formats.
- Also called the **Syntax Layer**.

Functions of Presentation Layer:

1. **Translation:** Converts data formats between sender and receiver.
2. **Encryption/Decryption:** Provides secure transmission.
3. **Compression:** Reduces number of bits for multimedia (text, audio, video).

7. Application Layer

- Closest to the end-user.
- Provides access to network services.
- Handles **network transparency, resource allocation**, etc.
- Not the applications themselves, but services supporting them.

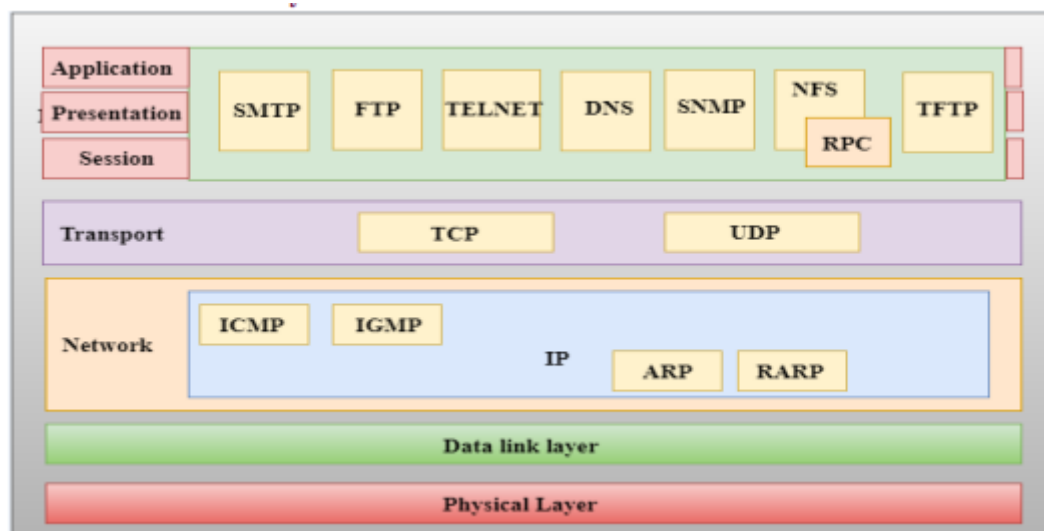
Functions of Application Layer:

1. **File Transfer, Access, and Management (FTAM):** Access, retrieve, and manage files remotely.
2. **Mail Services:** Provides email forwarding and storage.
3. **Directory Services:** Offers distributed database services and global information.

Summary

- **Computer Network** → interconnected devices for sharing data/resources.
- **OSI Model** → standard framework of **7 layers** that explain how communication occurs between systems, from physical transmission to user applications.

2. EXPLAIN TCP/IP PROTOCOL SUITE.



TCP/IP Protocol Suite

The **TCP/IP model** was developed **before the OSI model** and is the protocol suite on which today's internet runs.

It is not exactly the same as the OSI model, but both have similarities.

- TCP/IP consists of **five layers**:
 1. **Network Access Layer**
 2. **Internet Layer**
 3. **Transport Layer**
 4. **Application Layer***(Physical and Data Link are combined into the Network Access layer)*

1. Network Access Layer

- Lowest layer of TCP/IP.
- Combination of **Physical Layer + Data Link Layer** of OSI.
- Defines how data is sent physically through the network.
- Responsible for encapsulating IP datagrams into frames and mapping IP addresses to physical addresses.
- **Protocols:** Ethernet, Token Ring, FDDI, Frame Relay, X.25.

2. Internet Layer

- Also called the **Network Layer**.
- Main responsibility: **Send packets from source to destination**, regardless of the path taken.
- **Key Protocols:**
 - **IP (Internet Protocol):** Provides addressing, routing, packetizing, fragmentation, reassembly.
 - **ARP (Address Resolution Protocol):** Maps IP address to physical (MAC) address.
 - **ICMP (Internet Control Message Protocol):** Sends error messages and network diagnostics (e.g., ping).

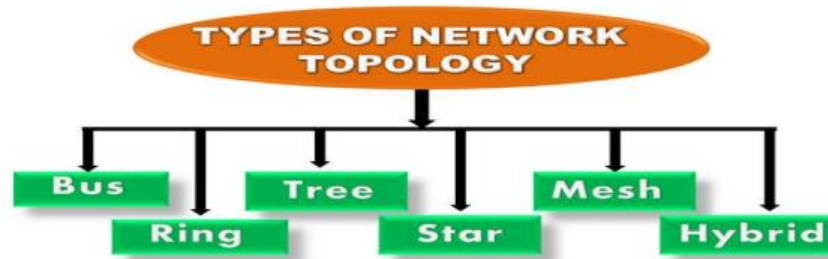
3. Transport Layer

- Provides **reliable data delivery** (error correction, sequencing, flow control).
- Main protocols:
 - **TCP (Transmission Control Protocol):**
 - Reliable, connection-oriented.
 - Divides data into segments, reorders at destination.
 - **UDP (User Datagram Protocol):**
 - Connectionless, unreliable but faster.
 - Suitable for streaming, gaming, VoIP.

4. Application Layer

- Topmost layer, provides **services directly to users and applications**.
- Handles representation and communication issues.
- **Protocols include:**
 - **HTTP/HTTPS** → Web browsing
 - **SMTP/IMAP/POP3** → Email
 - **DNS** → Domain name resolution
 - **FTP** → File transfer
 - **SNMP** → Network management
 - **Telnet/SSH** → Remote login
- **TCP/IP** is a hierarchical set of communication protocols.
- It has **five layers**, compared to **seven layers in OSI**.
- The internet is entirely based on **TCP/IP protocols**, which provide addressing, routing, error handling, and application services.

3. EXPLAIN DIFFERENT NETWORK TOPOLOGIES



Different Network Topologies

Topology defines the structure of a network — i.e., how devices (nodes) are physically or logically connected.

1. Point-to-Point Topology

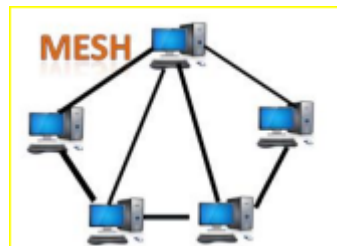
- Simplest form of topology.
- Two nodes are connected directly with a dedicated link.
- **Example:** Connection between a computer and a printer.

3. Bus Topology



- All nodes share a single communication line (backbone cable).
- Data sent by a node is received by all others, but only the addressed node accepts it.
- **Advantages:** Low cost, easy installation.
- **Disadvantages:** Signal interference, difficult troubleshooting, heavy cabling, reconfiguration issues.

4. Mesh Topology



- Each device is connected to multiple other devices with redundant paths.
- Provides high reliability and fast communication.
- **Types:**
 - **Full Mesh:** Every node connected to all others.
 - **Partial Mesh:** Only some nodes connected.
- **Advantages:** Reliable, fault-tolerant.
- **Disadvantages:** Expensive, complex cabling.
- **Example:** WANs, Internet.

5. Ring Topology



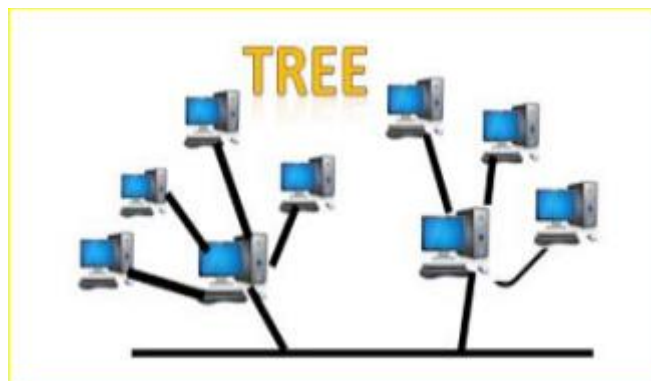
- Each node connects to exactly two others, forming a circular pathway.
- Data flows in one direction (unidirectional).
- Uses **token passing** to avoid collisions.
- **Advantages:** Reliable, inexpensive cabling.
- **Disadvantages:** Failure of one node can affect the whole network; difficult troubleshooting.

6. Star Topology



- All nodes are connected to a central hub/switch.
- The hub rebroadcasts data to all connected devices.
- **Advantages:** Easy troubleshooting, scalable, high speed.
- **Disadvantages:** If the hub fails, the whole network goes down.
- **Example:** Ethernet LANs.

7. Tree Topology

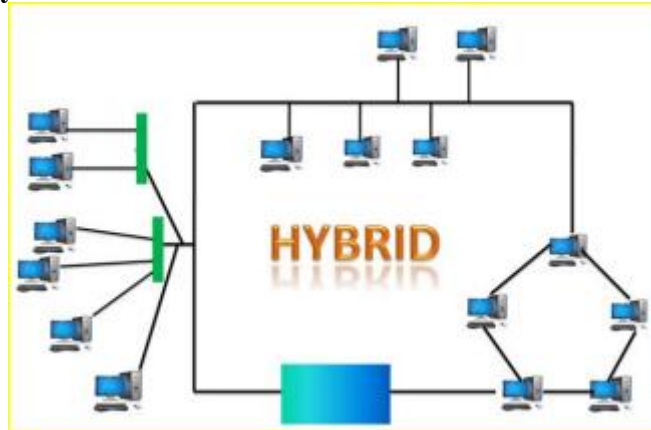


- Hierarchical structure (combination of star and bus).
- Root node connected to other nodes like a branching tree.
- **Advantages:** Easy expansion, supports broadband transmission.
- **Disadvantages:** Costly, difficult troubleshooting, relies on main bus cable.

7. Line Topology

- All nodes connected in a straight line.
- Simple but rarely used in modern networks.

8. Hybrid Topology



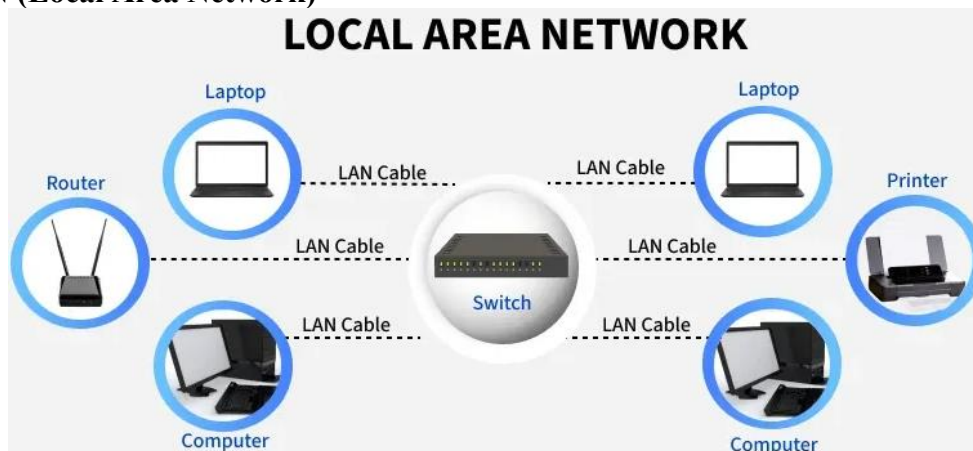
- Combination of two or more topologies (e.g., Star + Bus).
- **Advantages:** Flexible, reliable, scalable.
- **Disadvantages:** Complex design, high cost.
- **Example:** Banking networks combining multiple branch topologies.

Summary

- **Simple Topologies:** Point-to-Point, Line.
- **Popular in LANs:** Bus, Star, Ring.
- **Advanced:** Mesh, Tree, Hybrid

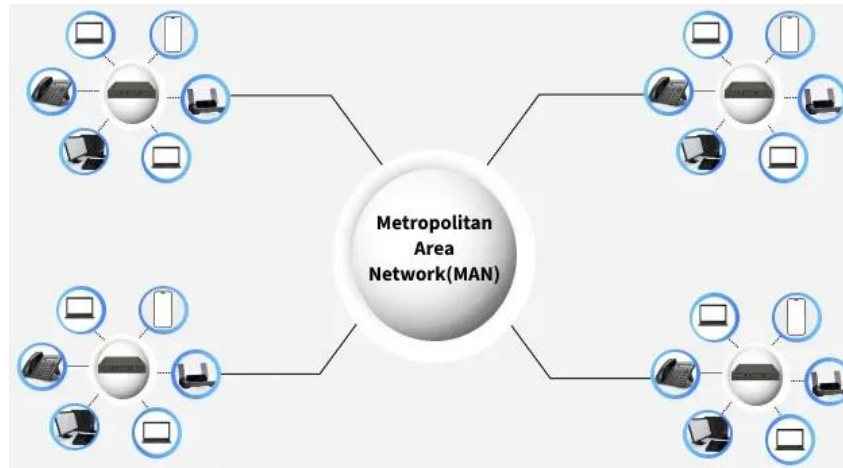
4. DIFFERENTIATE LAN MAN, WAN

1. LAN (Local Area Network)



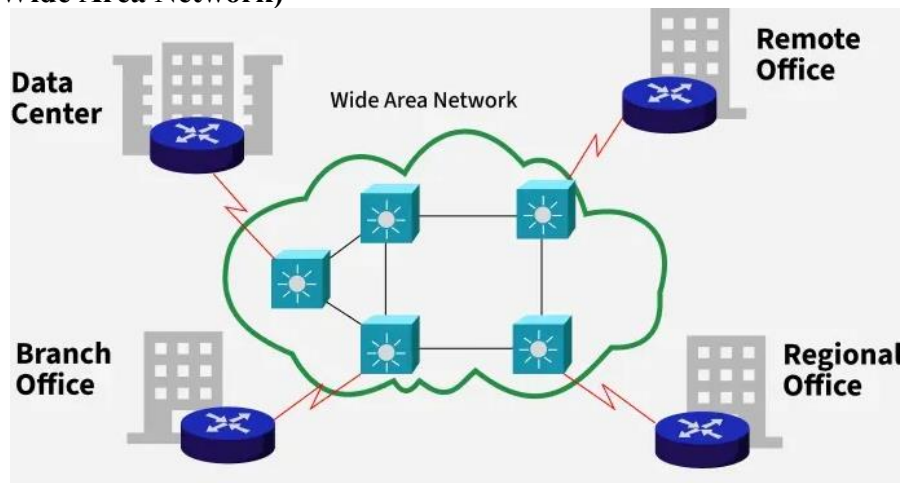
- Covers a **small geographical area** (e.g., building, office).
- **Inexpensive** to set up.
- Uses **Ethernet or Token Ring** technologies.
- Data transfer is **fast and secure**.
- Example: Computers connected in a school lab.

2. MAN (Metropolitan Area Network)



- Covers a **city or large campus**.
- Larger than LAN, smaller than WAN.
- Uses **Ethernet, ATM, Token Ring, FDDI**.
- Provides high-range connectivity.
- Example: Cable TV networks in a city.

3. WAN (Wide Area Network)



- Covers a **country or worldwide area**.
- Uses **telephone lines, satellite links, internet**.
- Internet is the **largest WAN**.
- Mostly used by **government and large organizations** to manage data.
- Example: The Internet.

Tabular Differentiation

Feature	LAN	MAN	WAN
Full Form	Local Area Network	Metropolitan Area Network	Wide Area Network
Coverage Area	Small area (building, office)	City or metropolitan area	Country or global
Technology	Ethernet, Token Ring	Ethernet, ATM, FDDI, Token Ring	Telephone lines, satellite, fiber
Speed	High (Fast, secure)	Moderate to high	Lower compared to LAN/MAN
Cost	Low	Medium	High
Example	Office network	Cable TV in a city	The Internet

5. EXPLAIN GUIDED MEDIA

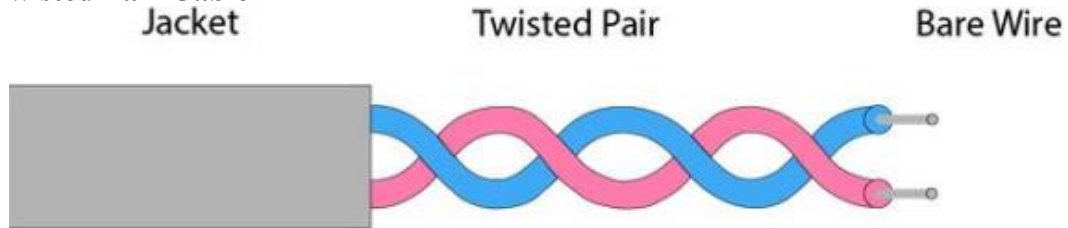
Guided Media

Guided media (also called **bounded media**) refers to the **physical medium** through which signals are transmitted.

These media use **cables/wires** to guide data signals from one device to another.

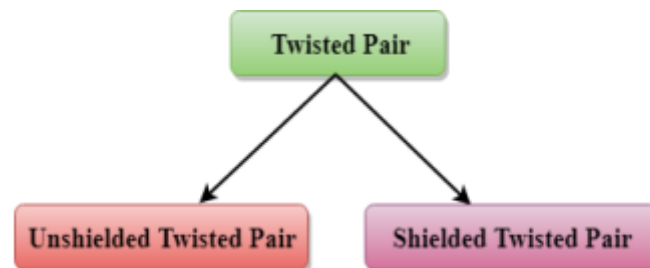
Types of Guided Media

1. Twisted Pair Cable



- Made of a pair of insulated copper wires twisted around each other.
- Inexpensive, lightweight, and easy to install.
- Frequency range: **0 to 3.5 KHz**.
- **Function:** Twisting reduces electromagnetic interference.

Types:



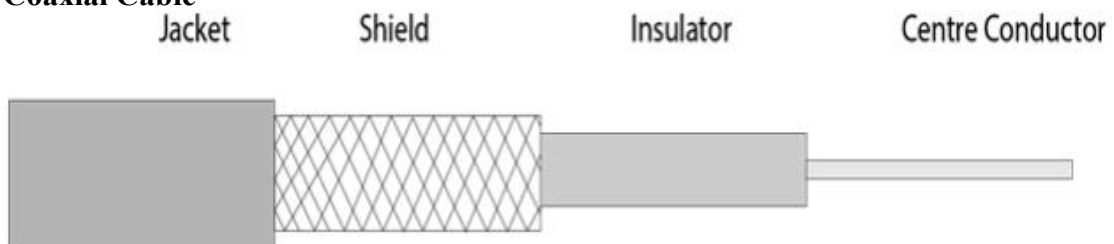
1. Unshielded Twisted Pair (UTP):

- Widely used in telecommunication.
- Categories:
 - Cat 1 → Telephone lines (low-speed data).
 - Cat 2 → Supports up to 4 Mbps.
 - Cat 3 → Supports up to 16 Mbps.
 - Cat 4 → Supports up to 20 Mbps (long-distance).
 - Cat 5 → Supports up to 200 Mbps.
- **Advantages:** Cheap, easy to install, used in LAN.
- **Disadvantage:** Short distance (attenuation).

2. Shielded Twisted Pair (STP):

- Contains shielding mesh around the wire for protection.
- Higher transmission rate than UTP.
- **Advantages:** Better performance, higher data rates.
- **Disadvantages:** More expensive, higher attenuation.

3. Coaxial Cable



- Commonly used in **TV wiring**.

- Has **two parallel conductors**:
 - Inner copper conductor
 - Outer copper mesh (shield)
- Middle core (non-conductive) separates them.
- Prevents **electromagnetic interference (EMI)**.

Types:

1. **Baseband Transmission:** Transmits a single signal at high speed.
2. **Broadband Transmission:** Transmits multiple signals simultaneously.

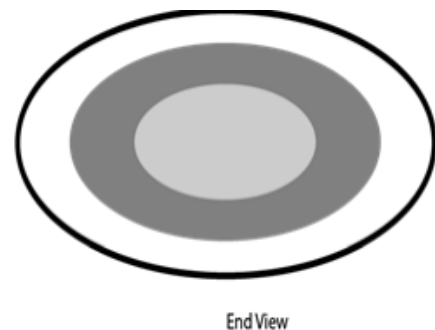
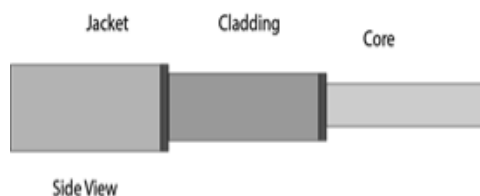
Advantages:

- High-speed data transfer.
- Better shielding than twisted pair.
- Higher bandwidth.

Disadvantages:

- More expensive than twisted pair.
- Faults can disrupt entire network.

3. **Optical Fiber (Fibre Optic Cable)**



- Uses **light signals** for communication instead of electrical signals.
- Consists of **optical fibers coated in plastic** for protection.
- Provides **higher speed and bandwidth** compared to copper wires.

Basic Elements:

1. **Core:** Central glass/plastic that transmits light.
2. **Cladding:** Surrounds core; reflects light back inside.
3. **Jacket:** Protective outer layer.

Advantages:

- Greater bandwidth (more data capacity).
- Faster transmission (light-based).
- Longer distances supported.
- More reliable (immune to interference).
- Thinner, lighter, stronger than copper cables.

Summary

- **Twisted Pair:** Cheap, short distances (used in LANs).
- **Coaxial:** Better shielding, used in TV & broadband.
- **Optical Fiber:** High bandwidth, long-distance, modern choice for fast internet.

6. EXPLAIN UNGUIDED MEDIA

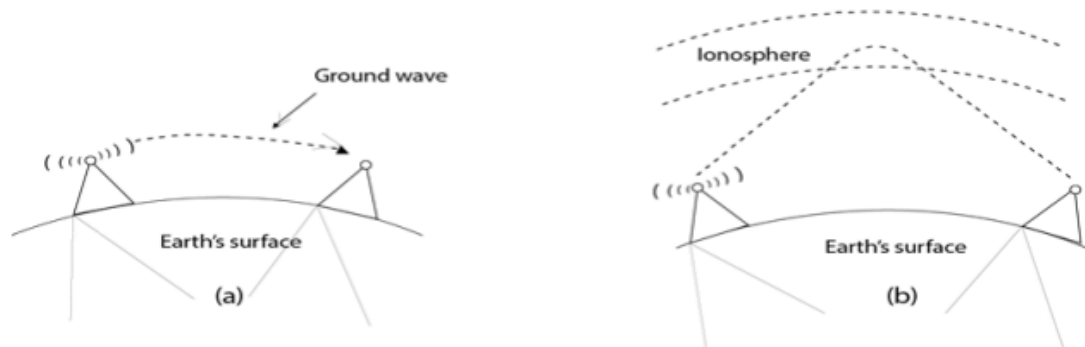
Unguided Media

Unguided media (also called **wireless transmission**) transmits **electromagnetic waves without any physical conductor**.

- The medium is **air**, where signals propagate freely.
- Commonly used for **wireless communication**.
- Types: **Radio Waves, Microwaves, Infrared**.

Types of Unguided Media

1. Radio Waves



- Electromagnetic waves transmitted in **all directions** (omnidirectional).
- Frequency range: **3 KHz to 1 GHz**.
- Sending and receiving antennas do not need alignment.
- **Examples:** FM radio, TV, cordless phones.

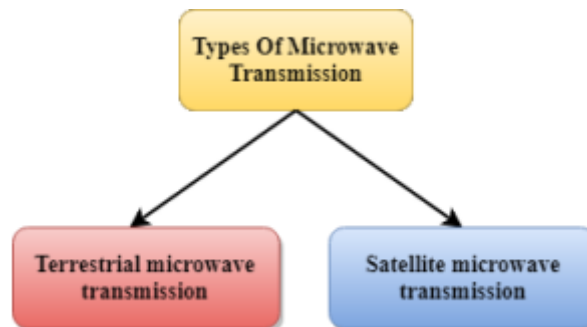
Applications:

- Useful for **multicasting** (one sender, many receivers).
- Used in **wide area networks (WANs)** and **mobile communication**.

Advantages:

- Covers large area.
- Can penetrate walls.
- Higher transmission rate.

2. Microwaves



Microwaves are **unidirectional** (require aligned antennas). Frequency range: **1 GHz to 1000 GHz**.

Types:

1. Terrestrial Microwave:

- Transmits focused beam from one ground antenna to another.
- Works on **line-of-sight** (antennas must face each other).
- Bandwidth: 1 to 10 Mbps.
- Used for communication across buildings or terrains.

Advantages: Cheaper than cables, works across oceans/terrains.

Disadvantages: Insecure (eavesdropping possible), weather-sensitive, limited bandwidth.

2. Satellite Microwave:

- A satellite relays signals between earth stations.
- Covers very large areas.
- Common in **TV broadcasting, weather forecasting, mobile communication**.

Advantages: Large coverage, independent of distance, easy installation.

Disadvantages: Very costly, limited life span (12–15 years), requires monitoring.

3. Infrared (IR)

- Frequency range: **300 GHz to 400 THz**.
- Used for **short-range communication** (within a closed area).
- Cannot penetrate walls → provides **security** from outside interference.

Applications:

- TV remotes, short-range file transfer between mobiles/computers.

Advantages:

- High bandwidth → high data rate.
- Secure (confined within a room).

Disadvantages:

- Useless outdoors (sunlight interferes).
- Limited to line-of-sight within short distance.

Summary

- **Radio waves:** Long-range, omnidirectional, used in WANs & broadcasting.
- **Microwaves:** Line-of-sight, used for terrestrial and satellite communication.
- **Infrared:** Short-range, secure, used in remotes and device-to-device transfer.

UNIT-II

1. EXPLAIN THE CRC TECHNIQUE WITH AN EXAMPLE.

VIDEO REFERENCE: <https://youtu.be/Q9G-oEYnAgE?si=RrGSZ77XA6A-TCoW>

NOTE: FOR OPENING VIDEOS PREFER MOBILE VIEW

Cyclic Redundancy Check (CRC)

Definition

Cyclic Redundancy Check (CRC) is an error detection technique used in the **Data Link Layer** of computer networks. It detects errors in the transmitted data by performing **binary division (modulo-2 division)** using a predefined generator polynomial (divisor).

- **Sender Side** → Appends CRC bits (remainder) to the data before transmission.
- **Receiver Side** → Divides received data by the same generator.
 - If remainder = 0 → No Error.
 - If remainder $\neq$ 0 → Error detected (retransmission requested).

Steps of CRC Technique

1. Select a **generator (divisor)** of size n bits.
2. Append $(n-1)$ zeros to the data (frame/message).
3. Perform **modulo-2 binary division** of the appended data by the generator.
4. Obtain the **remainder (CRC bits)**.
5. Append the remainder to the original data → **Codeword**.
6. At the receiver, divide the received codeword by the same generator:
 - If remainder = 0 → Correct data.
 - Else → Error detected.

Example Calculation (from video)

- **Message (Frame)** = 1101011011
- **Generator (Divisor)** = 10011 (5 bits)
- Degree = 4 → Append **4 zeros** to the message.

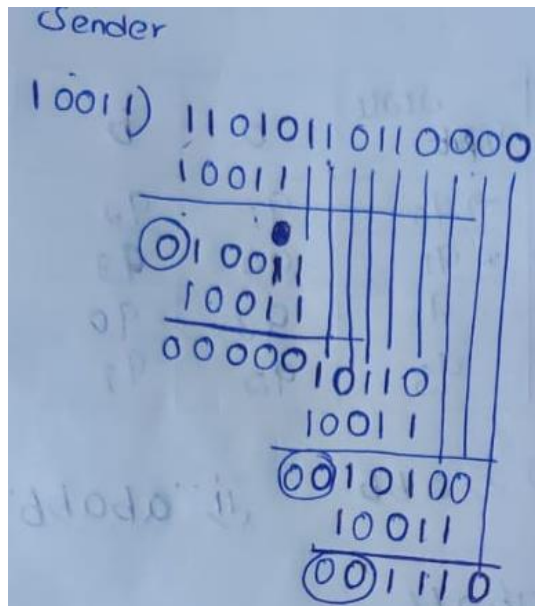
Step 1: Append Zeros

Dividend = 11010110110000

Divisor = 10011

Step 2: Perform Modulo-2 Division

We divide 11010110110000 by 10011 using XOR.



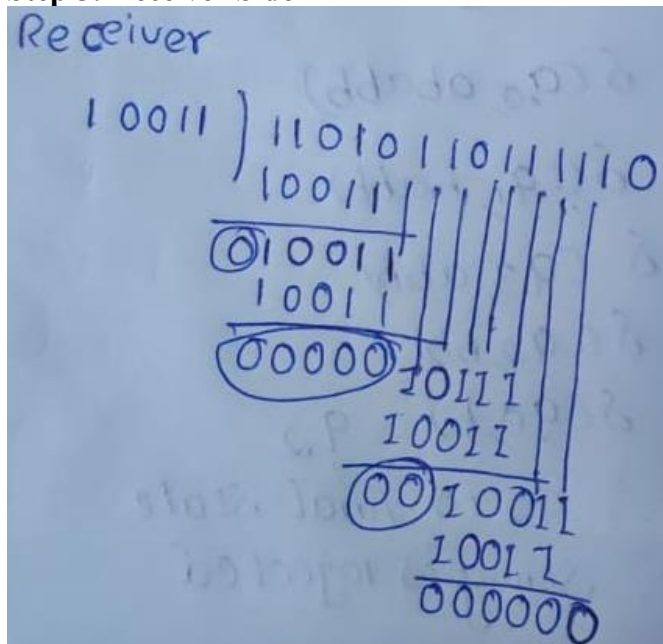
Step 3: Remainder (CRC bits)

Remainder = 1110

Step 4: Form the Codeword

$$\begin{aligned}\text{Codeword} &= \text{Message} + \text{Remainder} \\ &= 1101011011 + 1110 \\ &= 11010110111110\end{aligned}$$

Step 5: Receiver Side



- Receiver divides 11010110111110 by 10011.
- If remainder = 0000 → Message is **error-free**.
- If remainder $\neq 0$ → **Error detected**, request retransmission.

Key Points

- CRC detects **single-bit and burst errors** effectively.
- It **cannot correct errors**, only detect them.
- Widely used in **Ethernet, Wi-Fi, USB, HDLC, CAN bus**.

Final Answer

- **Data (Message)** = 1101011011

- **Generator (Divisor)** = 10011
- **Appended Zeros** = 0000
- **Remainder (CRC bits)** = 1110
- **Transmitted Codeword** = 11010110111110
- **Receiver Check** = Remainder 0000 → **No Error**

2. EXPLAIN CHECKSUM TECHNIQUE WITH A SUITABLE EXAMPLE.

VIDEO REFERENCE: <https://youtu.be/6ceMF-qcybI?si=J2LeZ2kdGshKA8pu>

Checksum Technique

Definition

The **Checksum** is an **error detection technique** used in the **Data Link Layer**.

- It is based on the concept of dividing the message into fixed-size segments, adding them together, and then taking the **complement** of the sum as the **checksum**.
- At the receiver side, all received segments (including the checksum) are added again. If the result is **all zeros**, the message is error-free; otherwise, an error is detected.

Steps of Checksum Technique

At Sender Side

1. **Divide the message** into n segments, each of size m bits.
2. **Add** all segments together using binary addition.
3. If a **carry** is generated from the addition, wrap it around and add it to the least significant bit (LSB).
4. **Complement** the final sum → This becomes the **Checksum**.
5. **Transmit** the original message + checksum to the receiver.

At Receiver Side

1. Receiver receives the **message + checksum**.
2. Adds all the segments (including the checksum).
3. If the final result = **all zeros**, the message is **accepted** (no error).
4. If the final result $\neq 0$, the message is **rejected** (error detected).

Example

10110011 | 10101011 | 01011010 | 11010101

Message size = **32 bits**

Divided into **4 segments of 8 bits each**:

Example for Checksum:

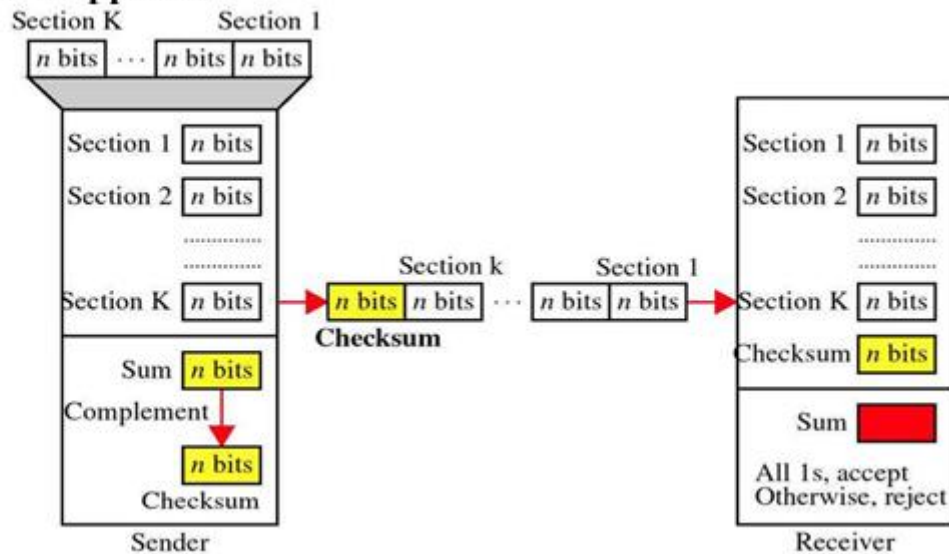
Example:
 $k=4, m=8$
 10110011
 10101011
 01011110
 11010101
 10001110
 Sum: 10001111
 Checksum 01110000

(a)

(a) Sender's end for the calculation of the checksum, (b) Receiving end for checking the checksum

Example: Received data
 10110011
 10101011
 01011110
 11010101
 10001110
 Sum: 11111111
 Complement = 00000000
 Conclusion = Accept data

(b)

Diagrammatic approach:**Receiver Side**

1. Receiver gets **4 message segments + checksum**.
2. Adds all 5 segments together:
3. Segment 1 + Segment 2 + Segment 3 + Segment 4 + Checksum
4. If result = 00000000 (all zeros), **No Error** → Accept message.
5. If result $\neq 0$, **Error Detected** → Request retransmission.

Result

- **Message Segments (4) + Checksum** = 00000000
- No error detected, message is correct.

Key Points

- Checksum is used for **error detection only** (not correction).
- It is simple and effective for detecting **single-bit errors and burst errors**.
- Widely used in **IP, TCP, UDP protocols**.

3. EXPLAIN FRAMING.**Framing in Computer Networks**

Framing is a function of the **Data Link Layer (DLL)**.

The physical layer only provides a raw bit stream (1s and 0s), but the data link layer must organize this stream into **discrete, manageable units** called **frames**.

Why Framing is Needed?

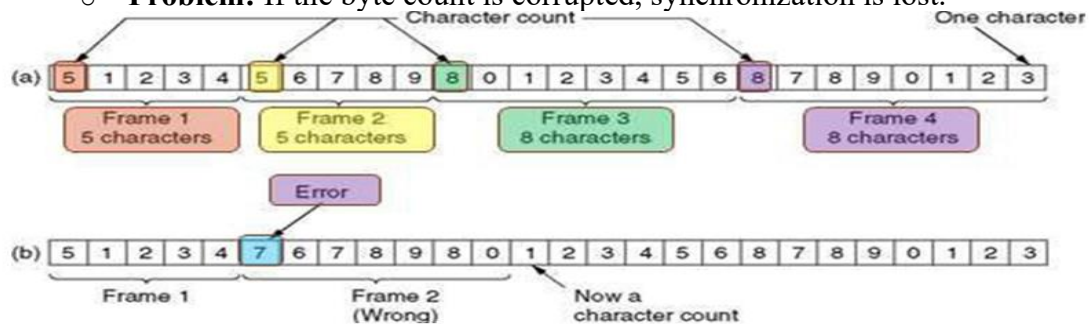
- To **separate messages**: The receiver should know where a frame starts and where it ends.
- To **detect errors**: Each frame has a **checksum** (error-detection code) so the receiver can check if the frame is corrupted.
- To **resynchronize**: If errors disturb the bit stream, the receiver must find the correct frame boundaries again.

Methods of Framing

There are four main techniques:

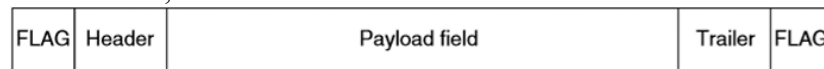
1. **Byte Count (Character Count)**
 - The first field of the frame header specifies the total number of bytes in the frame.

- Example: If the header says 8, the receiver knows the next 8 bytes belong to this frame.
- **Problem:** If the byte count is corrupted, synchronization is lost.

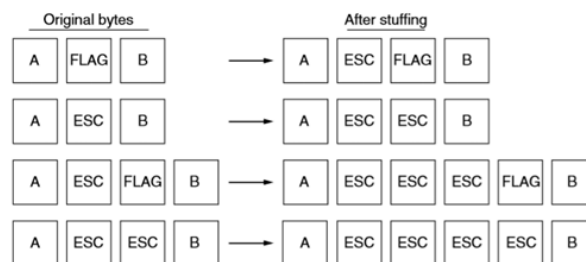


2. Flag Bytes with Byte Stuffing

- Each frame starts and ends with a special **flag byte** (e.g., 01111110).
- If the flag pattern accidentally appears in the data, an **escape character (ESC)** is inserted before it (byte stuffing).
- At the receiver side, ESC is removed.



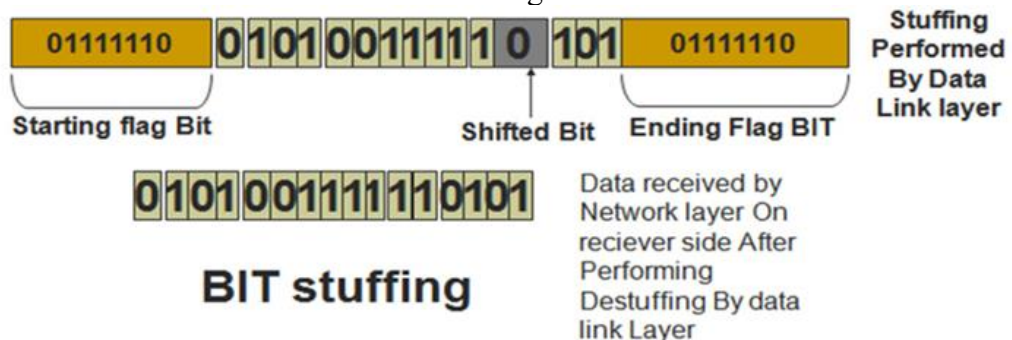
(a)



Four examples of byte sequences before and after byte stuffing.

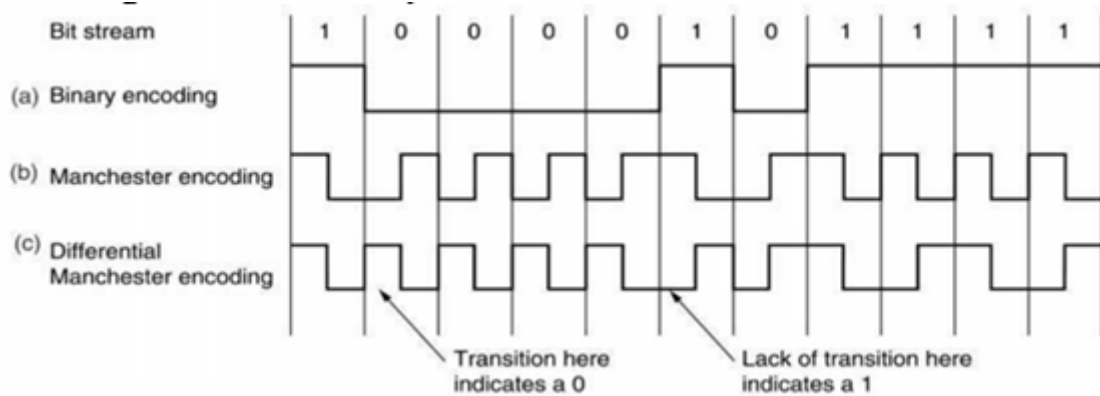
3. Flag Bits with Bit Stuffing

- Works at the **bit level** (used in **HDLC**).
- Each frame begins and ends with a unique **bit pattern** 01111110.
- If the sender's data has five consecutive 1s, it **automatically inserts a 0** (bit stuffing).
- Receiver removes the extra 0 when reading the frame.



4. Physical Layer Coding Violations

- Used in systems where the physical layer encoding has unused patterns.
- Example: In **Manchester encoding**, some bit combinations (like 11 or 00) are not normally used for data and can be reserved to mark frame boundaries.



Summary

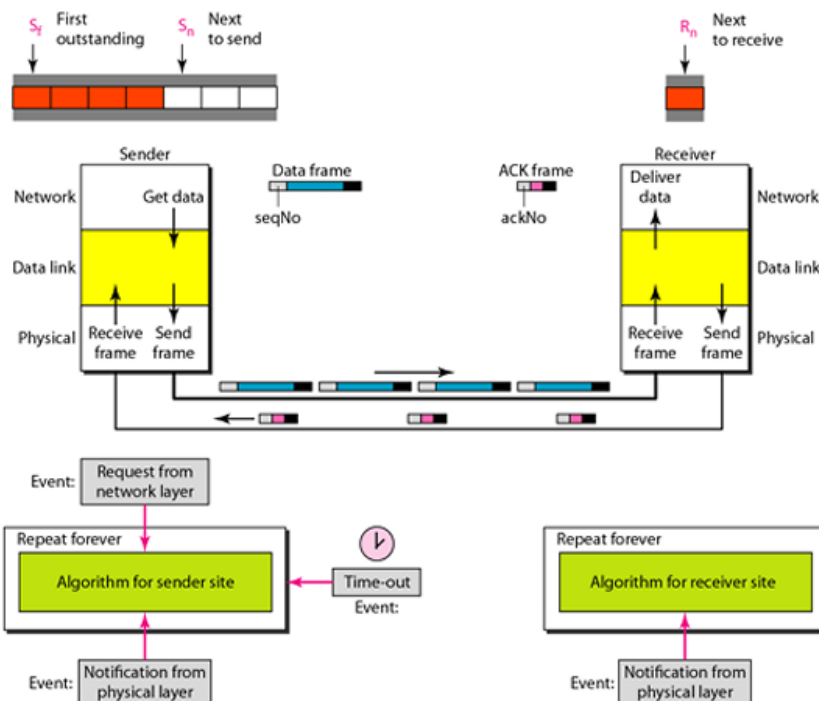
- **Framing = Splitting raw bit stream into frames.**
 - Each frame contains: **Header + Payload + Trailer (with error detection).**
 - **Techniques:**
 1. Byte count
 2. Flag bytes with byte stuffing
 3. Flag bits with bit stuffing
 4. Physical layer coding violations
- 4. EXPLAIN ABOUT THE GO-BACK N SLIDING WINDOW PROTOCOL AND DISCUSS ITS DRAW BACKS.**

VIDEO REFERENCE :

https://youtu.be/SmsJyMve43E?si=2kTnETTHHT_hmPCa

NOTE: DRAW THE DIAGRAM PROVIDED IN THE VIDEO

Go-Back-N Sliding Window Protocol



Go-Back-N ARQ Protocol

Definition

Go-Back-N ARQ (Automatic Repeat Request) is a **sliding window error control protocol** where:

- **Sender window size = N**

- **Receiver window size = 1**

This means the sender can transmit up to **N frames at a time** without waiting for acknowledgments, but the receiver can only accept the **next expected frame** in sequence.

The protocol uses **pipelining**, i.e., sending multiple frames from sender to receiver without waiting for ACKs of previous frames.

Working (Step-by-Step Example)

1. Initial Transmission

- Suppose the sender wants to send 7 frames (0 to 6).
- Sender window size = 4 → sender can send frames **0, 1, 2, 3** without waiting for ACKs.

2. ACK for Frame 0

- Receiver successfully receives frame 0 and sends ACK(0).
- Sender slides the window by 1 → window now covers frames **1 to 4**.
- Sender transmits frame 4.

3. ACK for Frame 1

- Receiver receives frame 1, sends ACK(1).
- Window slides → now covers **2 to 5**.
- Sender transmits frame 5.

4. Lost Acknowledgment (Case 1)

- Suppose receiver gets frame 2 and sends ACK(2), but ACK is **lost**.
- Sender waits for ACK until its timer (say 10 ms).
- On timeout → sender retransmits **frame 2 and all subsequent frames (3, 4, 5)** again.

5. Lost Frame (Case 2)

- If **frame 2 itself is lost**:
 - Receiver discards later frames (3, 4, 5) since they're out of order.
 - When sender's timer for frame 2 expires, it retransmits **frame 2 and all subsequent frames**.

Drawbacks of Go-Back-N ARQ

1. Wastage of Bandwidth

- Even if only one frame/ACK is lost, all subsequent frames are retransmitted unnecessarily.

2. Inefficient in Noisy Channels

- High error rates lead to repeated retransmissions, reducing throughput.

3. Receiver Buffering Not Utilized

- Receiver cannot store out-of-order frames (unlike Selective Repeat).

4. Long Recovery Time

- Delay increases since multiple frames must be retransmitted after a single error.

Go-Back-N ARQ improves efficiency over Stop-and-Wait ARQ by allowing multiple outstanding frames (pipelining). But, it suffers from **inefficiency due to unnecessary retransmissions** and works best in networks with **low error rates**.

5. EXPLAIN PPP

Point-to-Point Protocol (PPP)

PPP is a widely used **data link layer protocol** that provides a standard method for transmitting data over **point-to-point links** such as telephone lines, serial cables, ISPs, and DSL. Millions of Internet users historically used PPP to connect home computers to Internet Service Providers (ISPs).

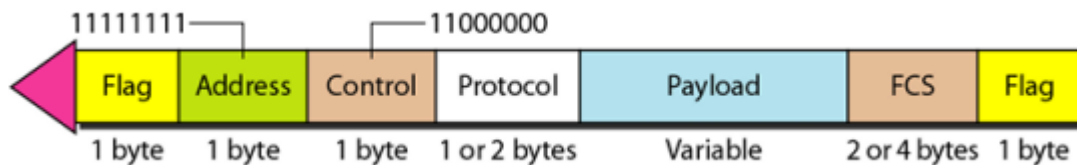
Services provided by PPP

1. Defines the **frame format** for data transmission.
2. Establishes and negotiates the **link connection** between two devices.
3. Encapsulates **network layer data** in data link frames.
4. Provides **authentication** between two devices (using PAP or CHAP).
5. Supports **multiple network layer protocols** (e.g., IP, IPX, AppleTalk).
6. Allows **multiple links** to be combined (Multilink PPP).
7. Provides **temporary network address configuration** (e.g., for dial-up users).

Limitations of PPP

- No **flow control** (sender can send continuously).
- Only **error detection** (CRC); no retransmission or correction.
- No **sophisticated addressing** (only point-to-point, not multipoint).

PPP Frame Format

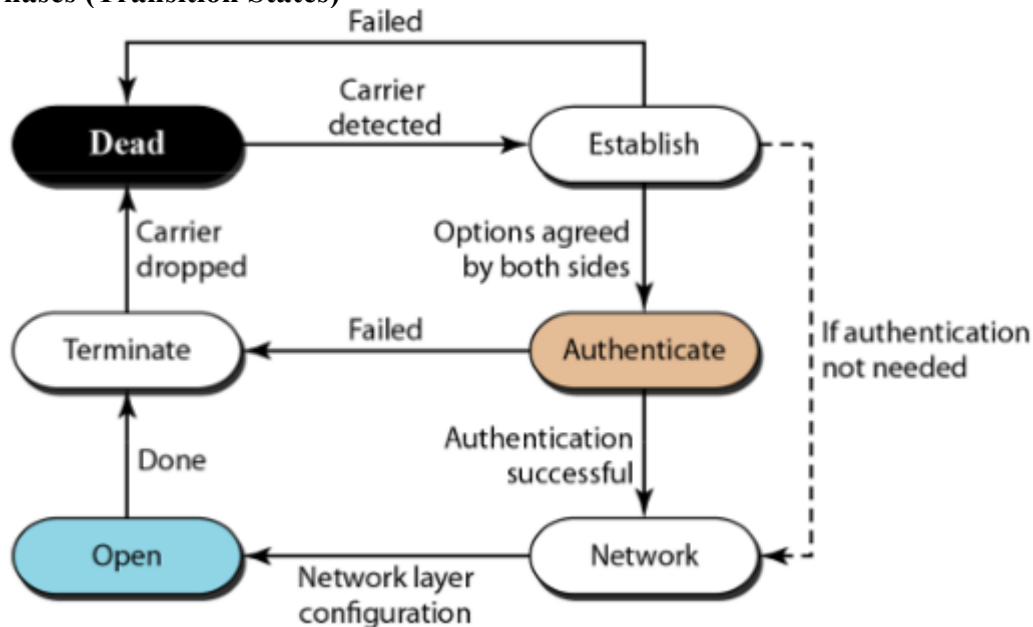


PPP is **byte-oriented** (unlike HDLC which is bit-oriented).

A PPP frame contains:

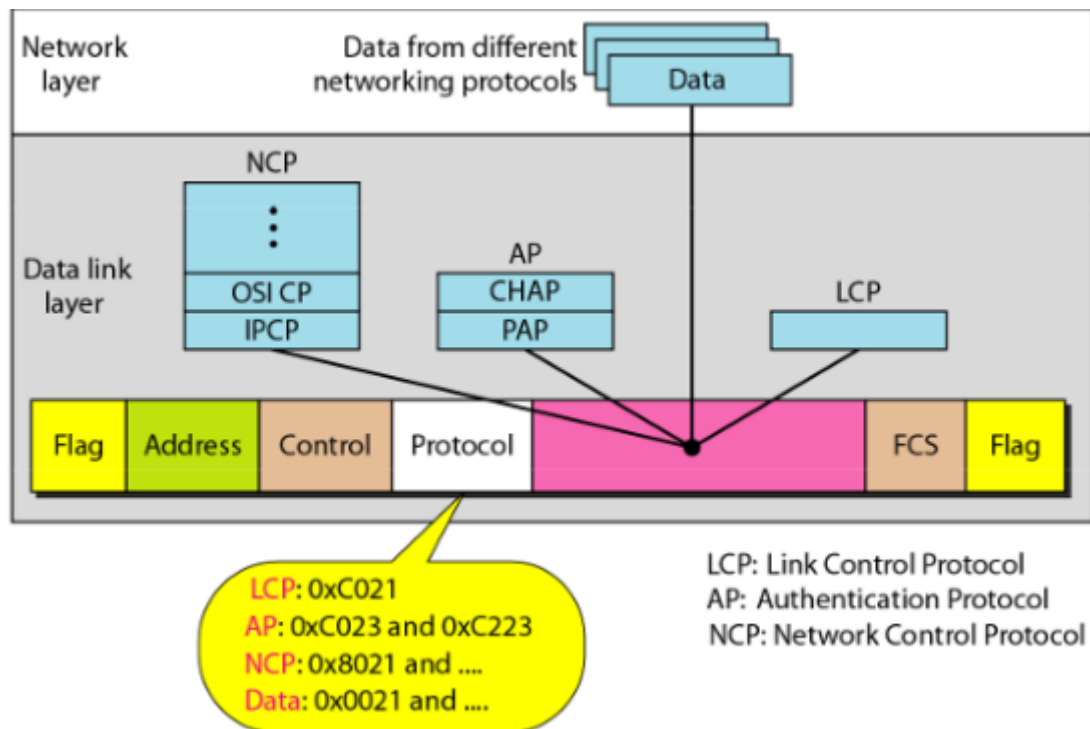
- **Flag (01111110):** Start and end of frame.
- **Address:** Always set to 11111111 (broadcast).
- **Control:** Constant 11000000 (no flow control).
- **Protocol:** Identifies payload (e.g., IP, LCP, NCP).
- **Payload:** Actual data (default max 1500 bytes).
- **FCS:** Error detection (CRC 16-bit or 32-bit).

PPP Phases (Transition States)



1. **Dead:** No link in use.
2. **Establish:** Negotiation of options (via LCP).
3. **Authenticate:** Optional; PAP or CHAP is used.
4. **Network:** Negotiate which network protocols to use (via NCP).
5. **Open:** Data transfer phase.
6. **Terminate:** Closing the link.

PPP Multiplexing



PPP uses:

- **LCP (Link Control Protocol):** Establishes/configures the link.
- **AP (Authentication Protocols):** PAP (Password Authentication Protocol) or CHAP (Challenge Handshake Authentication Protocol).
- **NCP (Network Control Protocols):** One for each network layer protocol (e.g., IPCP for IP).

Multilink PPP

- Allows **splitting one logical PPP frame** into **multiple physical frames** across different channels.
- Increases bandwidth by combining multiple links.

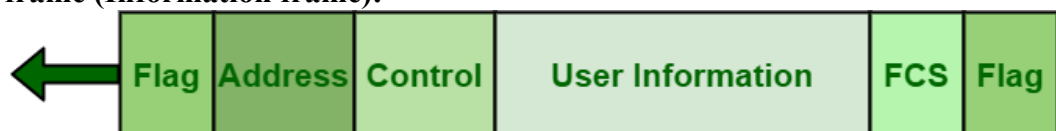
In short:

PPP is a simple, byte-oriented, point-to-point protocol that supports framing, authentication, and multiple network protocols but lacks flow control and sophisticated error correction.

6. LIST THE THREE TYPES OF HDLC FRAMES AND BRIEFLY EXPLAIN THEIR FUNCTIONS.

HDLC (High-Level Data Link Control) uses **three types of frames**, which are determined by the control field.

1. I-frame (Information frame):



- Used to carry **actual data/information** from the network layer.
- Includes both **send sequence number** and **acknowledgment number**.
- Supports piggybacking of acknowledgment in ABM (Asynchronous Balanced Mode).
- First bit of the control field is **0**.

2. S-frame (Supervisory frame):



- Used for **error control and flow control**.
- Contains only an **acknowledgment number** (no information field).
- Helps manage acknowledgments like **RR, RNR, REJ, SREJ**.
- First two bits of the control field are **10**.

3. **U-frame (Unnumbered frame):**



- Used for **link management and miscellaneous control functions** (e.g., setup and disconnection).
- Not sequenced and does not include acknowledgment numbers.
- Examples: **SNRM, SARM, SABM, UA, UP**.
- First two bits of the control field are **11**.